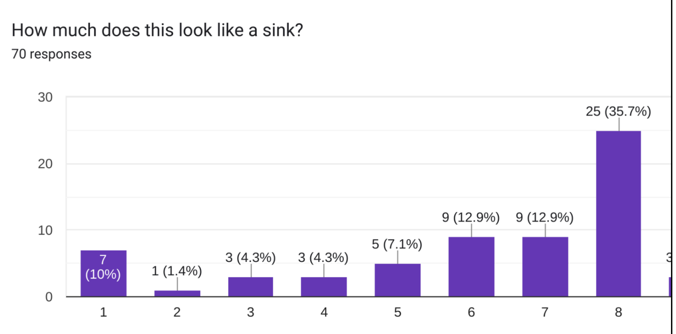
# Element I

**I1-I2: Importance and Comprehension of Testing Procedures**

| Design Requirements | Prio rity | Procedure/Results | Score |
|---|---|---|---|
| Produce Sufficient Water Flow | High (1) | After 1 minute of water flow, the measured amount of water that flowed out of the spigot was 0.4226753 gallons. This was measured with a beaker with the unit's millimeters. The measurement made resulted with 1600 millimeters, which is approximately .4226753 gallons. The data was the averages of 5 trials. | Satisfactory (2) |
| Portability | High (2) | The storage container weighs 7.8 pounds and the casing with the water containers inside weighed 17.6 pounds. Ultimately, all together the project (without any water) weighs 25.6 pounds. | Excellent (4) |
| Student Functionality | High (3) | To assess student functionality, we created a form where the stakeholder (Ms Brenan) provided her feedback. The results follow<br><br>90% of students were able to use the sink properly. | Good (3) |
| Standard Sink Appearance | High (4) | After conducting a poll of students within Plymouth Whitemarsh High School, this graph was produced. Mean score of 6.414 was produced.<br><br> | Satisfactory (2) |
| Safety | High (5) | There were 3 sharp edges recorded. This was reviewed and observed by all teammates. This gave | Satisfactory (2) |

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| | | a score of “Satisfactory”. | |
| Cleanliness | High (6) | After one week of use by the Whitemarsh Special Education Program, the recorded number of stains recorded on the Sink was _0___. This results with a score of ____. | Excellent (4) |
| Storage | Medi um (7) | Base:26.5x17.5x12.5   Top: 20x18x20 = 9<br><br>Overall: 14 × 22 × 32.5 = 10010 inches cubed=5.793ft cubed | Excellent (4) |
| Customizability | Low (8) | There were 2 observed features of the sink that were determined to be able to be customized by the teacher/student for different variations/ purposes. This included the amount of water that flows out, as well as the direction of the flow. This gave a score of “Good” | Good (3) |

**I3-I4: Detail and Summary of Data Analysis**

After completing our tests, we discovered that most of our criteria were met with a final rating of a “Good” or above. To begin, for our priority of sufficient water flow, we did not meet the requirement of “Good” however we were really close to the “Good” requirement. We were able to achieve a flow rate of 0.42 gallons per minute which after testing we determined that we might have been wrong on our judgment on what flow rate should be which score. We think that the perceived flow rate needed to receive a high score was highly overestimated as after using the sink, the results were off from our original thoughts. We determined the flow rate ranking should instead be excellent or close to as after conferring with experts from Drexel University and students testing the sink. The consensus was that the flow rate of the sink was well adapted and that if we had it more then we would run into issues with splashback onto the person using the sink. In our requirement for portability, we did really well when it came to the weight of the sink, only weighing 25.6 lbs, this met the goal of weighing under 50 lbs with an excellent rating by our matrix. This also complies with the Department of Labor, although this rating does not include the water added to the sink this is because the amount of water added can vary depending on what is being done with the sink and could always be lowered. We found that when filling both containers with water halfway, the weight would only be about a total of 40 lbs worth of water. While this is not as optimal for filling the containers up all the way, the overall weight of filling both containers fully would still put us in the range of “Good” on our matrix. For student functionality, we got a score of 90%. While we hoped to reach 100%, we believe this was acceptable, and hopefully could increase as the upcoming students at Whitemarsh enter the classroom. After completing a poll of students in our high school, we concluded that the average look of a sink was rated 6.414. This lands us a score of satisfactory on our matrix. This is not the best score as the sink not looking much like a sink could cause issues with students who have cognitive disabilities. However since this test has been completed we made adjustments to the look of the handles to be red and blue to match hot and cold. We also have added a mirror to the sink in order to simulate a bathroom sink in which a mirror is very common to be placed above.Next for the requirement of safety we measured by having all group members and other members in the class to touch and analyze the sink to see if they could find any sharp edges lying around. In total 3 were found this gave us a score of satisfactory. Although this score is not impressive, most of the sharp edges come from the top of the sink where not many students will be interacting with. Along with this because the sink is mostly made of wood and plastic, it would be possible to sand down these edges in order to make a smoother and safer surface around the sink. For cleanliness we scored a 4 which is excellent. We got this score by sending out a survey to the teacher who then went around the sink and looked for any scratches or dents and recorded them. Overall we are satisfied with this score as it shows that our sink can hold up under use for the life skills class at Whitemarsh Elementary .For storage we met the requirement that we set up, giving us a score of excellent. We were able to measure the volume of the sink overall to be 5.793ft cubed, this means that we have enough room to fit all of the waste water that would come out of water storage containers into the basin below. This would also allow the sink to be emptied not as quickly as if the storage was less volume. Finally for our requirement of customizability, all group members went around the outside of the sink and looked for spots where later improvements could be made such as adding a holder for a towel or more handles for the sink to be more portable. Overall 2 spots were determined and agreed upon to be places for later customization. This provided a score of "Good" , along with this the score could be improved if we were to make a larger sink however this clashes with portability as it becomes harder to move with a larger sink. To sum up, our average score across the board was a score of 3 (Good). We believe this was an acceptable score against our requirements, however there is room for improvement for future groups to achieve.